CLIP Embeddings as Product Characteristics in BLP Merger Simulation

Evidence from the hello→Colgate Acquisition in the US Toothpaste Market

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1 Data and Panel Construction

The dataset is an ASIN x quarter panel covering the Amazon US toothpaste market from 2018Q1 to 2022Q3 (19 quarters, 159 ASINs, 1,164 observations). Market shares are constructed using revealed-preference purchase data. Market size M_t is set to $(1+\lambda) \cdot \sum_j q_{jt}$ with $\lambda = 13.93$, implying an outside-good share of approximately 93%.

CLIP (Contrastive Language-Image Pre-training; Radford et al. 2021) embeddings are computed once per ASIN from its product image and listing text using `clip-vit-base-patch32`. Each modality's 512-dimensional embedding is first reduced to five modality-specific principal components; a joint PCA on the ten standardised columns (5 image + 5 text PCs) then yields five joint PCs (PC1–PC5) per ASIN. The joint PCA is fitted on the 155 ASINs with both modalities. Loadings are balanced across modalities — the image share of every joint component lies between 48.6% and 51.4% — confirming that neither modality dominates the product-space representation. Five clusters are obtained via K-means ($K = 5$, 50 restarts, seed 42), which explain 71.2% of the joint-PC variance.

Embeddings are static over the sample window. Established Amazon listings change their images and titles only marginally over 2018–2022, so re-extracting embeddings per quarter would produce near-identical vectors; the identifying variation is **cross-sectional** — ASINs within the same brand differ genuinely in packaging and benefit claims — and this is exactly the variation recovered by moving from brand-level to ASIN-level data. Time-varying demand conditions are absorbed by quarter fixed effects.

Twelve ASINs lack an image embedding (image retrieval failed at extraction time) but have valid text embeddings. Rather than dropping them or imputing brand means, they are projected into the joint PC space from their standardised text components, with image dimensions set to the cross-sectional mean (zero in standardised space). These ASINs span Crest (4), Sensodyne (2), Tom's of Maine (2), hello (2), and Other (2), and all clear the three-quarter inclusion filter. Dropping them would distort brand-level market shares and remove hello observations inside the merger simulation window; brand-mean imputation was rejected because it would reintroduce precisely the aggregation that the ASIN-level design is meant to avoid.

2 Demand Specification

2.1 Berry (1994) Linearisation

All three models are estimated via the Berry (1994) nested-logit linearisation:

$$\log \frac{s_{jt}}{s_{0t}} - \rho \, \log s_{j|g,t} = \alpha \, p_{jt} + \boldsymbol{\beta}' \mathbf{x}_{jt} + \mathrm{FE}_t + \xi_{jt}$$

where $s_{j|g,t} = s_{jt} / \sum_{k \in g} s_{kt}$ is the within-nest share, $\rho \in [0, 1)$ is the nesting intensity, and FE_t are quarter fixed effects (absorbed by within-quarter demeaning).

2.2 Price Coefficient Calibration

Valid cost-side instruments for price are unavailable in this high-frequency online retail dataset (see Section 5). Following the recommendation of Berry (1994) and standard practice in applied antitrust simulation, α is calibrated to match a target median own-price elasticity:

$$\alpha = \frac{\varepsilon_{\text{target}}}{\text{median}(p_{jt}(1 - s_{jt}))} = \frac{-2.62}{8.4502} = -0.310$$

The target $\varepsilon_{\text{target}} = -2.62$ corresponds to the brand-level CPG meta-analysis mean of Bijmolt, van Heerde & Pieters (2005), consistent with structural BLP estimates for analogous packaged-goods categories (Nevo 2001; Draganska & Jain 2006).

Sequential calibration note. Since α is calibrated at $\rho = 0$, the target elasticity -2.62 is the logit median. At the profiled nesting parameters ($\rho^* = 0.35$ for M2, $\rho^* = 0.67$ for M3), the nested-logit formula amplifies own-price sensitivity by approximately $1/(1 - \rho^*)$, yielding implied median elasticities of approximately -4.0 (M2) and -7.9 (M3).

2.3 ρ^* Selection — Concentrated OLS Profile

For each ρ on a fine grid $\{0, 0.01, \ldots, 0.90\}$, the adjusted dependent variable is formed and $\boldsymbol{\beta}(\rho)$ is estimated by OLS. The profiled ρ^* minimises the residual sum of squares:

$$\tilde{y}_{jt}(\rho) = \log \frac{s_{jt}}{s_{0t}} - \alpha p_{jt} - \rho \log s_{j|g,t}, \qquad \rho^* = \arg\min_{\rho} \left\| \tilde{y}(\rho) - \mathbf{X}\hat{\boldsymbol{\beta}}(\rho) \right\|^2$$

Table 1: Summary Statistics by CLIP Cluster

			Price (USD)			Mkt. Share (x1000)					
Cluster	Dom. Brand	N ASINs	$\bar{p}$	$\tilde{p}$	σ_p	$\bar{s} \times 10^3$	$\tilde{s} \times 10^3$	$\sigma_s \times 10^3$	$\bar{w}$	$\tilde{w}$	σ_w
C1	Sensodyne	29	10.26	9.63	5.58	1.068	0.770	0.848	249	249	145
C2	Colgate	33	8.00	8.39	3.65	1.303	0.817	1.259	537	399	296
C3	Crest	29	8.72	6.88	4.18	1.267	0.757	1.282	375	431	128
C4	Tom's of Maine	13	9.85	9.76	2.96	1.170	0.770	0.972	374	370	94
C5	Other (hello)	55	9.46	7.86	6.84	0.839	0.670	0.736	171	141	105

Notes:

Panel: 1,164 ASIN x quarter observations, 19 quarters (2018Q1–2022Q3). Mean/Median/SD of price in USD, market share scaled x1000, packa

3 Tables

Table 2: Demand Parameter Estimates — Three Models

Variable	M1: Logit (ρ=0)	M2: Brand-Nested (ρ^*=0.35)	M3: CLIP-Nested (ρ^*=0.67)
α (price)	**−0.310 (a)**	**−0.310 (a)**	**−0.310 (a)**
Package size (g)	0.0037*** (0.0003)	0.0037*** (0.0003)	0.0034*** (0.0003)
CLIP PC_1	-0.4034*** (0.0424)	-0.2869*** (0.0418)	-0.3280*** (0.0407)
CLIP PC_2	0.2033*** (0.0374)	0.2681*** (0.0368)	0.2103*** (0.0359)
CLIP PC_3	-0.1667*** (0.0386)	-0.1479*** (0.0380)	-0.2854*** (0.0370)
CLIP PC_4	0.2115*** (0.0399)	0.1574*** (0.0393)	0.2707*** (0.0382)
CLIP PC_5	-0.1652*** (0.0390)	-0.1243*** (0.0384)	-0.1745*** (0.0374)
Other brand	0.0849 (0.1327)	0.4590*** (0.1307)	0.2641** (0.1272)
Import x freight shock	0.6767*** (0.2342)	0.6709*** (0.2306)	0.7461*** (0.2245)
ρ^*	0.000 (b)	0.350 (b)	0.670 (b)
Implied median ε	-2.62	≈ -4.0	≈ -7.9
RSS drop vs ρ=0	—	3.1%	8.1%

Notes:

*** p<0.01, ** p<0.05, * p<0.10. FEs by quarter. N=1,164. (a) calibrated alpha; (b) grid-searched rho*. Secs. 2.2-2.3.

Table 3: Merger Counterfactuals: hello-to-Colgate Acquisition (2020Q1)

Brand	Obs. p	MC	Nash (pre)	Nash (post)	Δp	Δp %
M1: Logit (ρ=0)						
hello	**6.43**	**3.20**	**7.10**	**7.16**	**0.064**	**1.233%**
Colgate	**8.45**	**5.16**	**8.29**	**8.30**	**0.007**	**0.104%**
Tom's of Maine	10.47	7.17	10.70	10.71	0.006	0.069%
Crest	8.87	5.60	9.18	9.18	0.000	0.000%
Sensodyne	10.45	7.17	10.05	10.05	0.000	0.000%
M2: Brand-Nested (ρ^*=0.35)						
hello	**6.43**	**3.20**	**7.10**	**7.17**	**0.065**	**1.275%**
Colgate	**8.45**	**5.16**	**8.30**	**8.30**	**0.006**	**0.104%**
Tom's of Maine	10.47	7.17	10.71	10.71	0.006	0.069%
Crest	8.87	5.60	9.18	9.18	0.000	0.000%
Sensodyne	10.45	7.17	10.07	10.07	0.000	0.000%
M3: CLIP-Nested (ρ^*=0.67)						
hello	**6.43**	**5.20**	**7.05**	**7.07**	**0.020**	**0.404%**
Colgate	**8.45**	**5.88**	**8.59**	**8.59**	**0.002**	**0.018%**
Tom's of Maine	10.47	7.19	10.72	10.72	0.002	0.021%
Crest	8.87	5.63	9.18	9.18	0.000	0.000%
Sensodyne	10.45	7.28	10.29	10.29	0.000	0.000%

Notes:

All prices USD; post-merger averages (2020Q1–2022Q3). Bold = merger parties. Nash (pre): Bertrand-Nash under pre-merger ownership. Nash

The central economic-plausibility comparison across the three models is summarised below. The share of observations with **negative implied marginal costs** is the key diagnostic: when the demand system understates substitutability, the Bertrand inversion attributes implausibly large markups, pushing recovered costs below zero. CLIP-based nesting reduces the negative-MC rate from 8.6% to 1.7% — a five-fold improvement over both the plain logit and brand nesting — while simultaneously delivering the largest RSS improvement.

Table 4: Cross-Model Comparison: Demand Fit and Economic Plausibility

Model	ρ^*	Med. ε_{jj}	RSS drop	Neg. MC %	Med. Lerner	hello Δp	Colgate Δp
M1: Logit	0.00	-2.62	—	8.6%	38.5%	+1.23%	+0.104%
M2: Brand-Nested	0.35	-3.83	3.1%	8.6%	38.5%	+1.27%	+0.104%
M3: CLIP-Nested	**0.67**	**-7.31**	**8.1%**	**1.7%**	**28.9%**	**+0.40%**	**+0.018%**

Notes:

All statistics over the full estimation panel (1,164 ASIN x quarter observations) except merger effects (post-merger quarters 2020Q1-

4 Figures

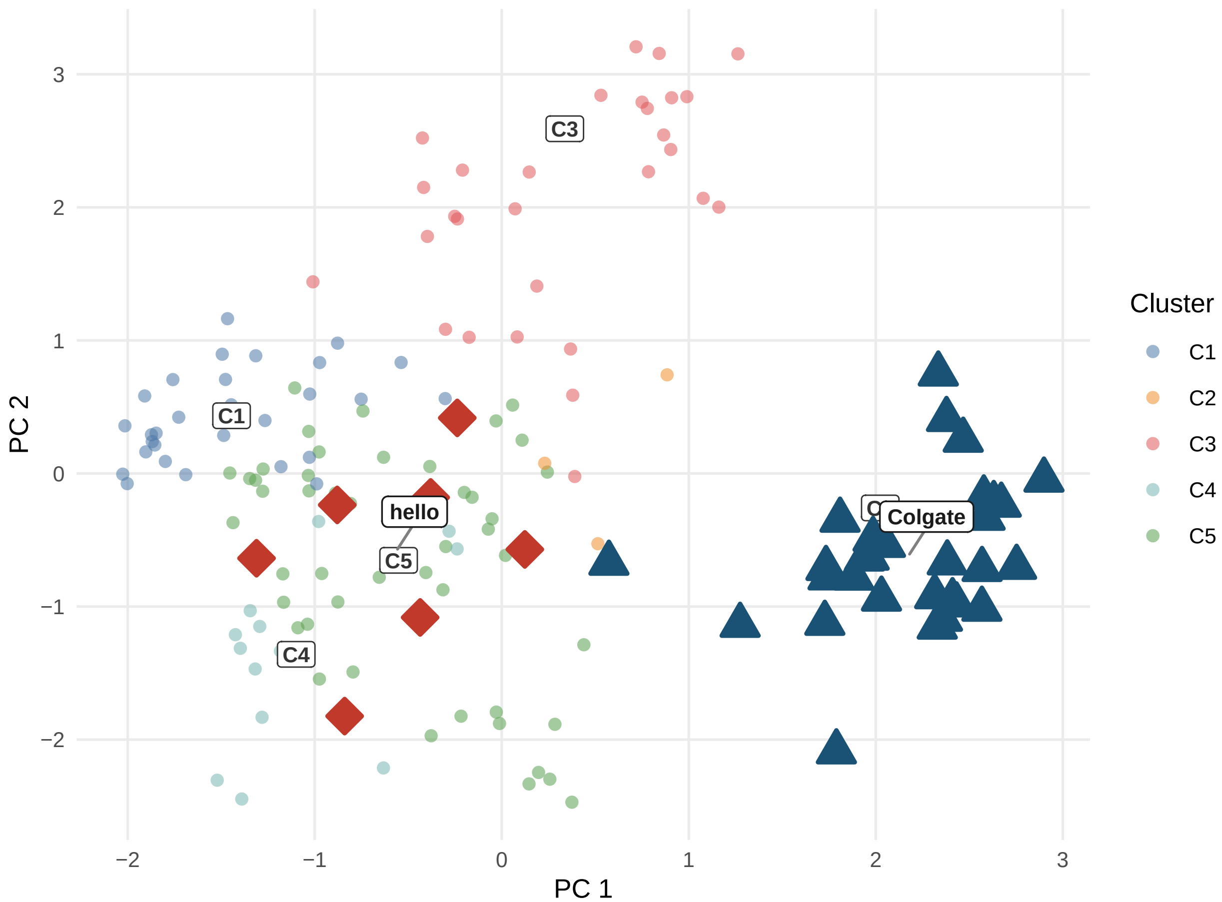

Figure 1: Product space from CLIP embeddings. Each point is one ASIN positioned by its first two principal components of the joint image+text embedding. Cluster centroids are labelled C1–C5. hello (red diamonds, C5) and Colgate (dark blue triangles, C2) occupy different clusters, implying a lower diversion ratio and a smaller predicted merger effect under CLIP nesting relative to plain logit.

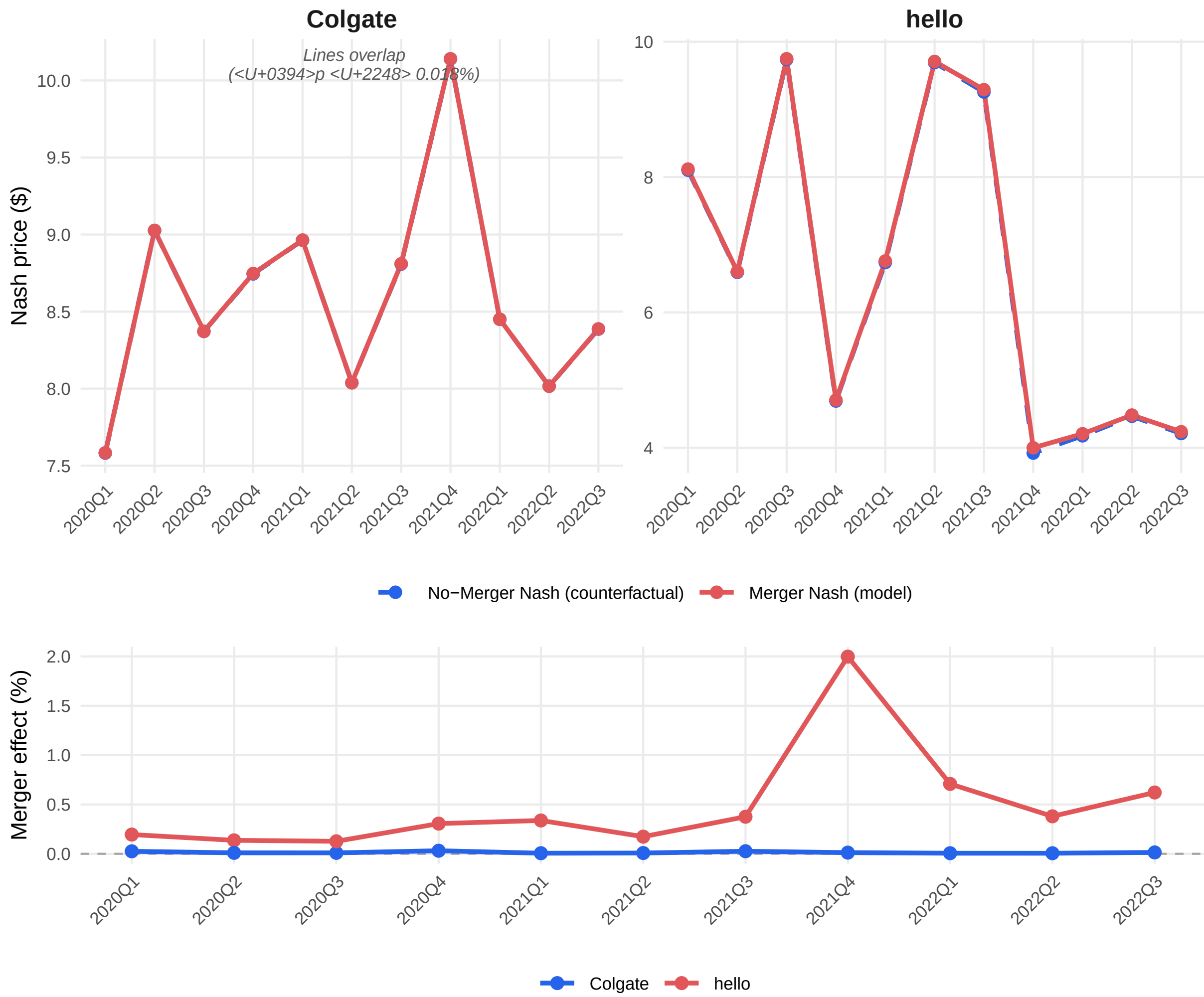

Figure 2: Merger simulation price trajectories (Model 3, CLIP-Nested, $\rho^* = 0.67$). Top panel: Nash equilibrium prices under merged and pre-merger ownership. Observed prices are omitted from the top panel because their quarter-to-quarter variation (\$4–\$10) would compress the counterfactual divergence. Bottom panel: merger price effect in percent, $(p^{\text{merger}} - p^{\text{no-merger}})/p^{\text{no-merger}} \times 100$. hello (red) shows a positive and time-varying effect; Colgate (blue) remains near zero.

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5 Bertrand–Nash Merger Simulation

The Bertrand–Nash merger simulation follows the standard approach in Berry (1994) and Nevo (2000). Marginal costs are recovered from the pre-merger first-order condition:

$$\mathbf{mc} = \mathbf{p} + (\Omega_{\text{pre}} \circ \Delta)^{-1}\mathbf{s}$$

where $\Omega[j,k] = \mathbf{1}[\text{firm}_j = \text{firm}_k]$ and Δ is the demand Jacobian with elements:

$$\Delta_{jj} = \alpha s_j \left[\tfrac{1}{1-\rho} - \tfrac{\rho}{1-\rho} s_{j|g} - s_j\right], \quad \Delta_{jk}^{\text{same}} = \alpha s_j \left[-\tfrac{\rho}{1-\rho} s_{k|g} - s_k\right], \quad \Delta_{jk}^{\text{diff}} = -\alpha s_j s_k$$

Marginal costs are averaged per ASIN across pre-merger quarters and held fixed. Two Nash equilibria are computed via the damped fixed-point iteration of Nevo (2000):

$$\mathbf{p}_{n+1} = \lambda \left[\mathbf{mc} - (\Omega \circ \Delta(\mathbf{p}_n))^{-1} \mathbf{s}(\mathbf{p}_n)\right] + (1-\lambda)\, \mathbf{p}_n$$

with $\lambda = 0.4$, convergence tolerance 10^{-8}, and up to 2,000 iterations. The post-merger simulation replaces Ω_{pre} with Ω_{post} (hello ASINs assigned to Colgate's firm identifier from 2020Q1). The merger price effect is the difference between these two equilibria: $\Delta p = p^{\text{merger}} - p^{\text{no-merger}}$.

Algorithm validation. The manual R implementation is validated against PyBLP's `compute_prices()` (Conlon & Gortmaker 2020). For both M1 (plain logit) and M3 (CLIP-nested, $\rho^* = 0.67$), the maximum absolute difference across 847 post-merger observations is below 2×10^{-8}, confirming algorithmic equivalence.

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6 References

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7 Appendix

7.1 A: K Sensitivity Analysis

Table 5: K Sensitivity Analysis: CLIP Cluster Count K = 2–10

K	WCSS	Silhouette	ρ^*	RSS Drop	Neg. MC%	hello $\Delta p\%$	Colgate $\Delta p\%$
2	1042.0	0.199	0.60	6.1%	0.5%	1.71%	0.015%
3	836.7	0.256	0.66	7.8%	0.9%	3.49%	0.013%
4	641.2	0.330	0.57	5.7%	3.0%	3.96%	0.041%
5	**456.5**	**0.388**	**0.67**	**8.1%**	**1.7%**	**0.40%**	**0.018%**
6	365.6	0.387	0.60	6.7%	3.4%	0.59%	0.044%
7	322.0	0.368	0.48	4.6%	4.4%	0.69%	0.055%
8	279.3	0.400	0.37	2.9%	4.7%	0.82%	0.067%
9	247.7	0.403	0.35	2.4%	4.7%	1.26%	0.177%
10	230.7	0.373	0.40	3.8%	4.6%	0.79%	0.065%

Notes: K-means on 5 CLIP PCs (seed 42, 50 restarts). WCSS = within-cluster sum of squares. Silhouette = average silhouette width (higher = better separation). ρ^* = argmin RSS on grid [0, 0.01, ..., 0.90]. RSS drop = $100 \times (RSS_0 - RSS^*)/RSS_0$. Neg. MC = share of ASINs with negative implied marginal cost. hello / Colgate Δp = average post-merger price effect in percent (2020Q1–2022Q3). **Bold/green row = selected specification (K=5).**

7.2 B: Model Fit — RMSE and MAE

Table 6: Model Fit: RMSE / MAE vs. Observed Prices (Post-Merger Quarters 2020Q1–2022Q3)

	RMSE / MAE (USD)		
Brand / Scope	M1: Logit	M2: Brand-Nested	M3: CLIP-Nested
All	**1.558 / 0.743**	**1.556 / 0.746**	**1.564 / 0.810**
hello	1.579 / 0.836	1.580 / 0.837	1.533 / 0.799
Colgate	1.144 / 0.559	1.142 / 0.560	1.149 / 0.652

Notes: Each cell shows RMSE / MAE in USD against observed transaction prices. Nash (merger) prices are used as the model prediction. “All” = all 847 ASIN × quarter observations in the post-merger period (2020Q1–2022Q3); hello = 35 obs.; Colgate = 172 obs. Pre-merger marginal costs are held fixed in both Nash simulations.